Discrete and Continuous Distributions

NAT

today

CLO

1. CLO 1: APPLY the concept of probability in relation to the concept of sampling error. (C3)
2. CLO 2: EXPLAIN the concept of point and confidence interval estimation for given parameters. (C2, A4)
3. CLO 3: DESIGN appropriate hypothesis testing based on given statistical problems. (C5, P7)

Objectives

1. Parametric Distributions
2. Calculate probability of discrete distributions: Uniform, Bernoulli, Binomial and Geometric, Poisson, Hypergeometric
3. Calculate probability of continuoud distributions

Discrete and Continuous Random Variable

Random variables can be categorized into discrete, continuous and mixed random variables. We are focussing on the first two. This chapter will explore the parametric distributions.

Discrete Random Variables

- Uniform (discrete) Distribution
- Bernoulli Distribution
- Binomial Distribution
- Poisson Distribution

Uniform (discrete) Distribution

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A random variable $X$ follows the discrete uniform distribu

$$f(x) = \frac{1}{N}$ for $x=1,2,..,N$$

where $N$ is the parameterss
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Continuous Distributions

As in the case of discrete variables, continuous random variables may be described by known probability density functions (PDFs).

Continuous Uniform Distribution $U(a, b)$

A continuous random variable X has a Uniform distribution if the probability of any possible value occurring is the same over the interval $[a, b]$.

- **PDF**: $f(x) = \frac{1}{b-a}$ for $a \leq x \leq b$
- **Mean**: $\mu = \frac{a+b}{2}$
- **Variance**: $\sigma^2 = \frac{(b-a)^2}{12}$

Exponential Distribution $Exp(\theta)$

Normally associated with "waiting-time" between events. It is memoryless.

- **PDF**: $f(x) = \frac{1}{\theta} e^{-x/\theta}$ for $x > 0$
- **Mean**: $\mu = \theta$ (Note: $\theta = 1/\lambda$)